

Lab Report: EEE 4878

Group No #10

Lab No #01

LAB TITLE:

Study of the basic current-versus voltage curve, threshold voltage, transconductance and figure of merit of the MOS transistor in the process technology library (PDK)

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1 ABSTRACT

This experiment was conducted to study the current-voltage characteristics, threshold voltage, transconductance, output resistance, and figure of merit of MOS transistors using the Cadence Virtuoso simulation environment and gpd090 technology library. DC sweep analyses were performed to obtain the I_D - V_{GS} and I_D - V_{DS} characteristics of NMOS and PMOS transistors under different biasing conditions and process corners. The threshold voltage was extracted using the constant current method, while transconductance (g_m), output resistance (r_o), and transconductance efficiency (g_m/I_D) were analyzed from simulated curves. The experiment also included comparison of NMOS and PMOS mobility for equal transistor dimensions. Results showed that NMOS mobility is higher than PMOS mobility, threshold voltage slightly varies across process corners, g_m increases with V_{GS} , and r_o decreases as the transistor moves toward strong inversion. The g_m/I_D curve demonstrated high efficiency in weak inversion and gradual reduction in strong inversion. These observations matched theoretical expectations and provided practical insight into MOSFET operation and analog circuit design.

2 KEYWORDS

1. MOSFET
2. Cadence
3. Threshold
4. Transconductance
5. Mobility

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3 INTRODUCTION

This experiment focused on studying the electrical characteristics of MOS transistors using the Cadence Virtuoso simulation platform and gpdk090 technology library. The behavior of NMOS and PMOS devices under different biasing conditions was investigated through DC analysis.

The objective of this experiment was to understand important transistor parameters such as threshold voltage, mobility ratio, transconductance (g_m), output resistance (r_o), and transconductance efficiency (g_m/I_D). The effect of different process corners on device performance was also analyzed.

During the experiment, several plots including I_D - V_{GS} , I_D - V_{DS} , g_m - V_{GS} , r_o - V_{GS} , and g_m/I_D - V_{GS} were obtained and analyzed. Threshold voltage extraction and mobility comparison between NMOS and PMOS transistors were also performed.

Understanding MOS transistor characteristics is essential in analog and digital integrated circuit design. The results of this experiment help designers predict transistor behavior under different operating conditions and process variations, which is important for designing reliable VLSI systems.

4 THEORY

The drain current equations of a MOSFET are given by:

4.1 Saturation Region

$$I_D = \frac{1}{2}\mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad (1)$$

4.2 Linear Region

$$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right] \quad (2)$$

where μ is carrier mobility, C_{ox} is oxide capacitance, W/L is aspect ratio, V_{GS} is gate-to-source voltage, V_{DS} is drain-to-source voltage, and V_{TH} is threshold voltage.

The transconductance is defined as:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad (3)$$

The output resistance is defined as:

$$r_o = \frac{\partial V_{DS}}{\partial I_D} \quad (4)$$

The transconductance efficiency is:

$$\frac{g_m}{I_D} \quad (5)$$

5 PRE-WORK DATA AND RESULTS

No pre-work was assigned for this experiment.

6 LAB HANDOUT QUESTIONS

- (1) For same W and L , i.e. $W = 480nm$ and $L = 100nm$, find the mobility ratio of NMOS and PMOS, i.e. μ_n/μ_p , at $(V_{GS} = 0.6V, V_{DS} = 0.8V)$, $(V_{GS} = 0.5V, V_{DS} = 1V)$, and $(V_{GS} = 1.2V, V_{DS} = 1.2V)$. Give comments on the values obtained.
- (2) What value of threshold voltage have you obtained in Part 7 for NMOS and PMOS? Give your comments on that.
- (3) Plot r_o vs V_{GS} of the NMOS transistor and comment on the r_o values at different operating conditions.
- (4) Give your comment on the g_m vs V_{GS} curve obtained in Part 8(1).
- (5) Give your comment on the g_m/I_D curve obtained in Part 8(2).

7 TOOLS USED

1. Cadence Virtuoso
2. gpdk090 Process Design Kit (PDK)
3. Spectre Simulator
4. Cadence Calculator Tool

8 PROCEDURE

1. An NMOS transistor schematic was created in Cadence Virtuoso using the gpdk090 library.
2. DC sweep analysis was performed by varying V_{GS} while keeping V_{DS} constant to obtain the I_D - V_{GS} characteristics.
3. Another DC sweep was carried out by varying V_{DS} for different values of V_{GS} to obtain the I_D - V_{DS} characteristics.

4. PMOS transistor simulations were also performed using the same dimensions to compare carrier mobility.
5. Threshold voltage was extracted using the constant current method from the I_D - V_{GS} curve.
6. The transconductance (g_m) and output resistance (r_o) were calculated using the Cadence calculator tool.
7. The g_m/I_D curve was obtained by dividing transconductance by drain current.
8. Simulations were repeated for different process corners such as SS, FF, SF, and FS to observe process variation effects.

9 RESULTS

9.1 Mobility Ratio of NMOS and PMOS

For same transistor dimensions,

$$W = 480nm, \quad L = 100nm$$

The mobility ratio was calculated from the drain current ratio of NMOS and PMOS:

$$\frac{\mu_n}{\mu_p} = \frac{I_{Dn}}{I_{Dp}} \quad (6)$$

9.1.1 Schematic

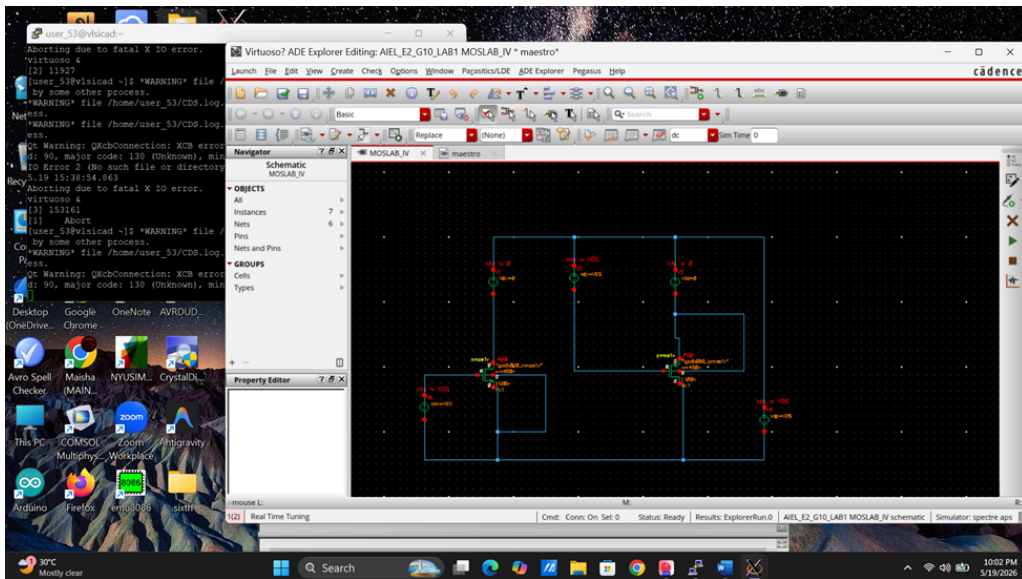


Figure 1: Schematic for mobility ratio calculation of NMOS and PMOS.

9.1.2 Result for $V_{GS} = 0.6V$, $V_{DS} = 0.8V$

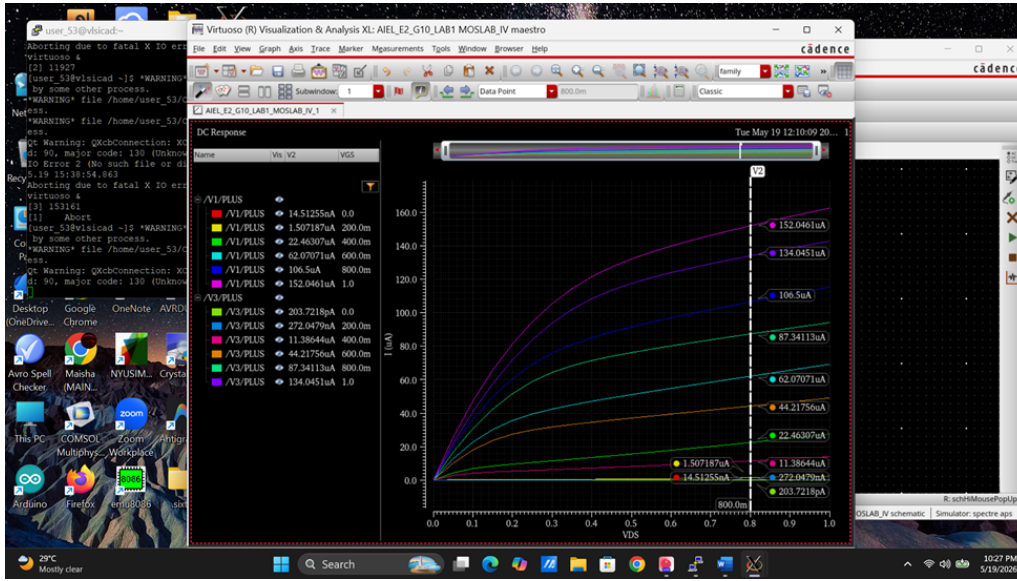


Figure 2: I_D - V_{DS} curve for $V_{GS} = 0.6V$ and $V_{DS} = 0.8V$.

From the graph,

$$I_{Dn} = 62.07071\mu A$$

$$I_{Dp} = 44.21756\mu A$$

$$\frac{\mu_n}{\mu_p} = \frac{62.07071}{44.21756} = 1.404$$

9.1.3 Result for $V_{GS} = 0.5V$, $V_{DS} = 1V$

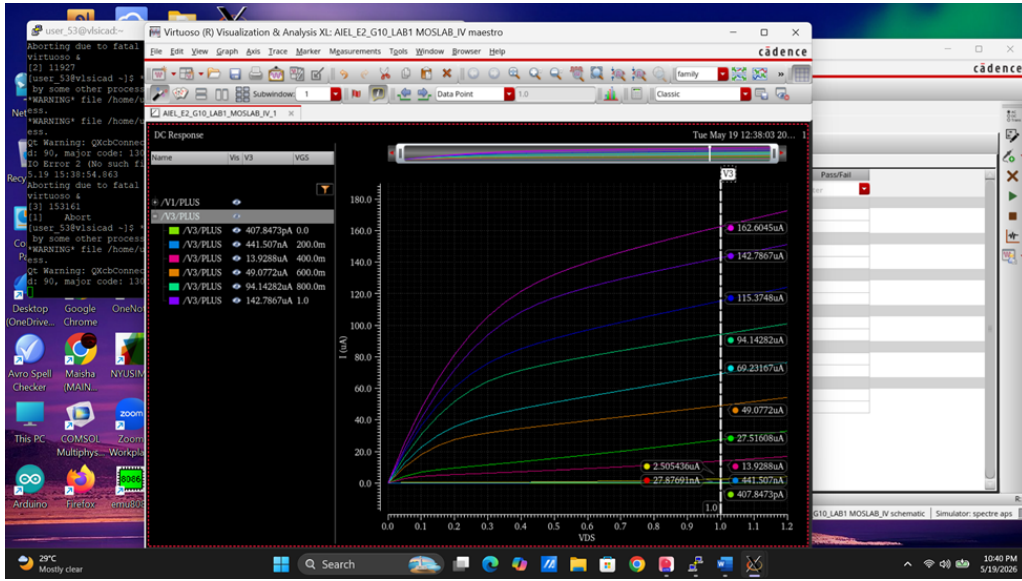


Figure 3: I_D - V_{DS} curve for $V_{GS} = 0.5V$ and $V_{DS} = 1V$.

From the graph,

$$I_{Dn} = 69.23167 \mu A$$

$$I_{Dp} = 49.0772 \mu A$$

$$\frac{\mu_n}{\mu_p} = \frac{69.23167}{49.0772} = 1.411$$

9.1.4 Result for $V_{GS} = 1.2V$, $V_{DS} = 1.2V$



Figure 4: I_D - V_{DS} curve for $V_{GS} = 1.2V$ and $V_{DS} = 1.2V$.

From the graph,

$$I_{Dn} = 172.5808\mu A$$

$$I_{Dp} = 151.1457\mu A$$

$$\frac{\mu_n}{\mu_p} = \frac{172.5808}{151.1457} = 1.142$$

9.1.5 Comment

The mobility ratio μ_n/μ_p was obtained from the drain current ratio of NMOS and PMOS transistors having $W = 480nm$, $L = 100nm$. The calculated values are approximately 1.404, 1.411, and 1.142 for the three bias conditions. Since all values are greater than 1, it can be concluded that electron mobility in NMOS is higher than hole mobility in PMOS. However, the ratio is not constant for all biasing conditions because effective mobility depends on the operating voltage. At higher V_{GS} and V_{DS} , mobility ratio decreases due to mobility degradation, velocity saturation, and stronger vertical electric field.

9.2 Threshold Voltage of NMOS and PMOS

The threshold voltage was extracted from the I_D vs V_{GS} curve using the constant current method.

9.2.1 NMOS Schematic

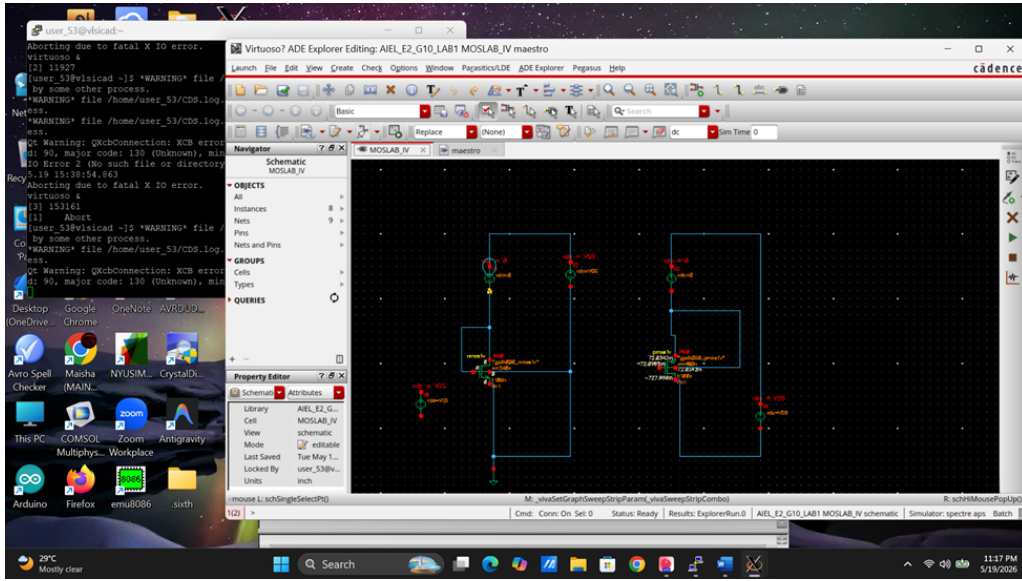


Figure 5: NMOS schematic for threshold voltage extraction.

9.2.2 NMOS Threshold Voltage Result

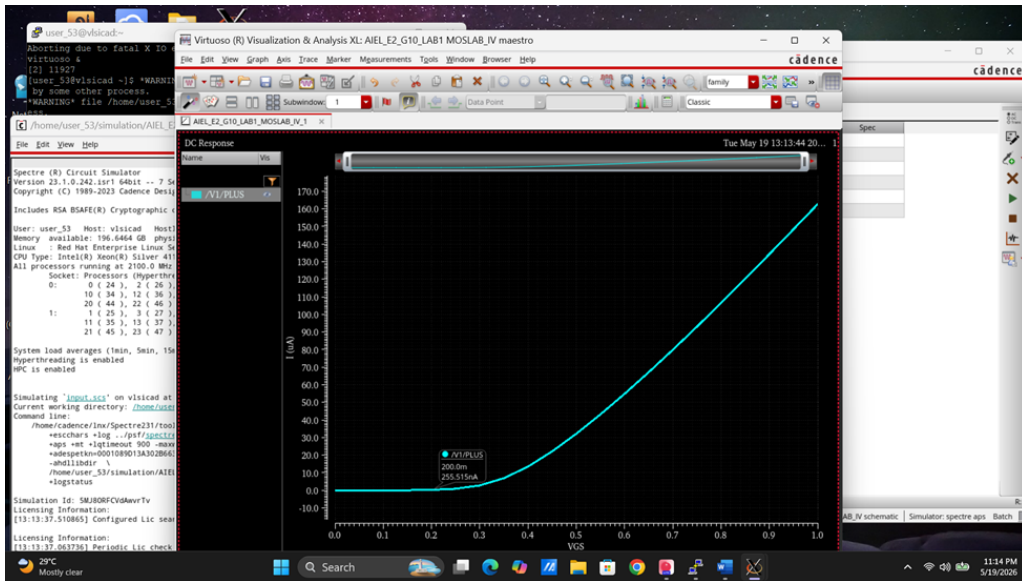


Figure 6: I_D vs V_{GS} curve for NMOS threshold voltage extraction.

From the NMOS curve,

$$V_{TH,n} \approx 0.200V$$

At this point,

$$I_D \approx 253.515nA$$

9.2.3 PMOS Schematic

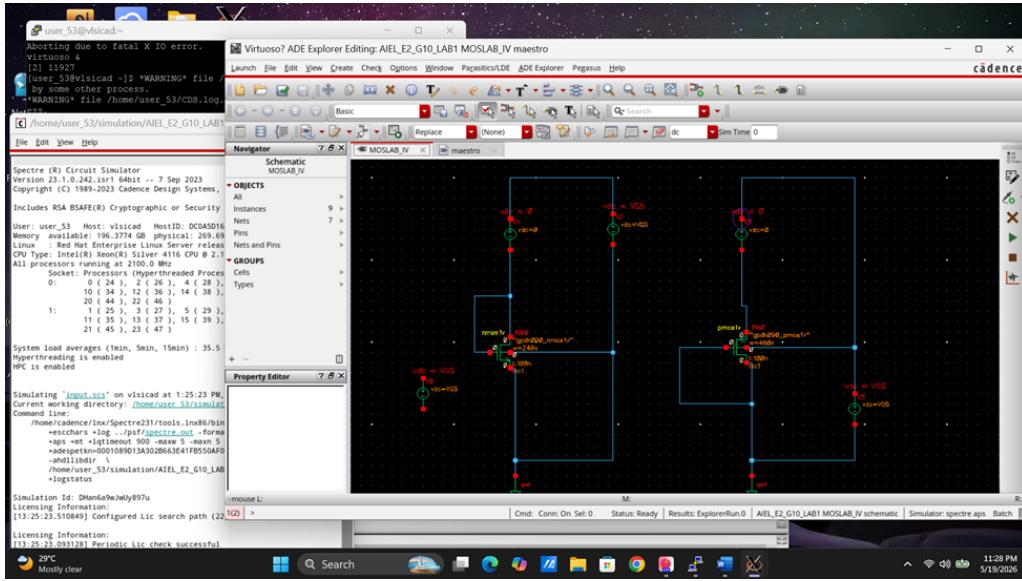


Figure 7: PMOS schematic for threshold voltage extraction.

9.2.4 PMOS Threshold Voltage Result

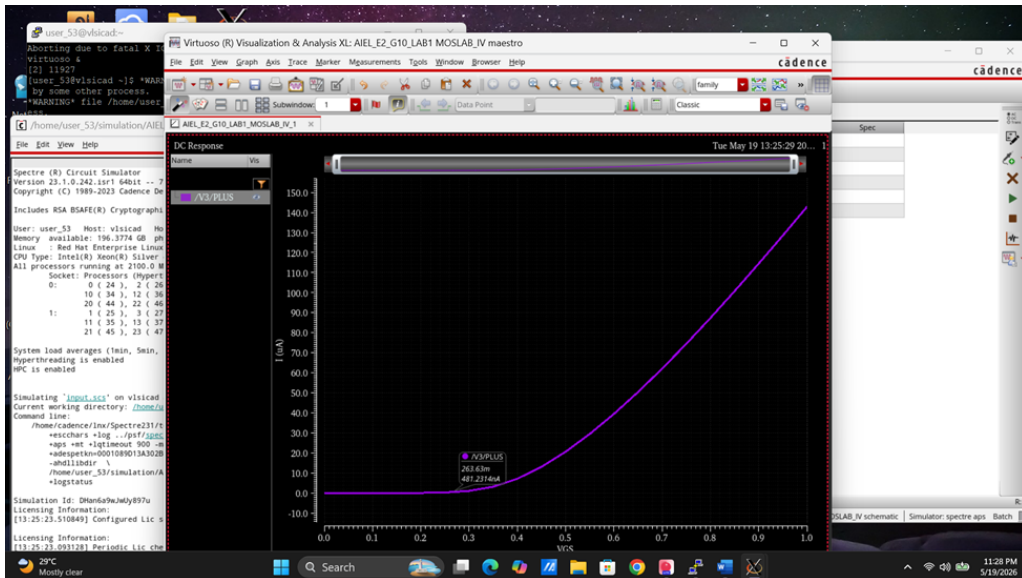


Figure 8: I_D vs V_{GS} curve for PMOS threshold voltage extraction.

From the PMOS curve,

$$|V_{TH,p}| \approx 0.26363V$$

At this point,

$$I_D \approx 487.237nA$$

9.2.5 Comment

The extracted threshold voltage of the NMOS transistor is approximately $0.200V$, while the magnitude of the PMOS threshold voltage is approximately $0.26363V$. The PMOS threshold voltage is higher in magnitude than the NMOS threshold voltage. For NMOS majority carriers are electrons and for PMOS majority carriers are holes. As hole mobility is lower than electron mobility PMOS devices require a slightly higher threshold voltage to achieve comparable conduction.

9.3 r_o vs V_{GS} of NMOS Transistor

The output resistance of the NMOS transistor was plotted with respect to gate-to-source voltage V_{GS} . The output resistance was obtained using:

$$r_o = \frac{V_{DS}}{I_D} \quad (7)$$

Here, V_{DS} was kept constant at $0.5V$ while V_{GS} was swept.

9.3.1 Result at $V_{GS} = 50mV$

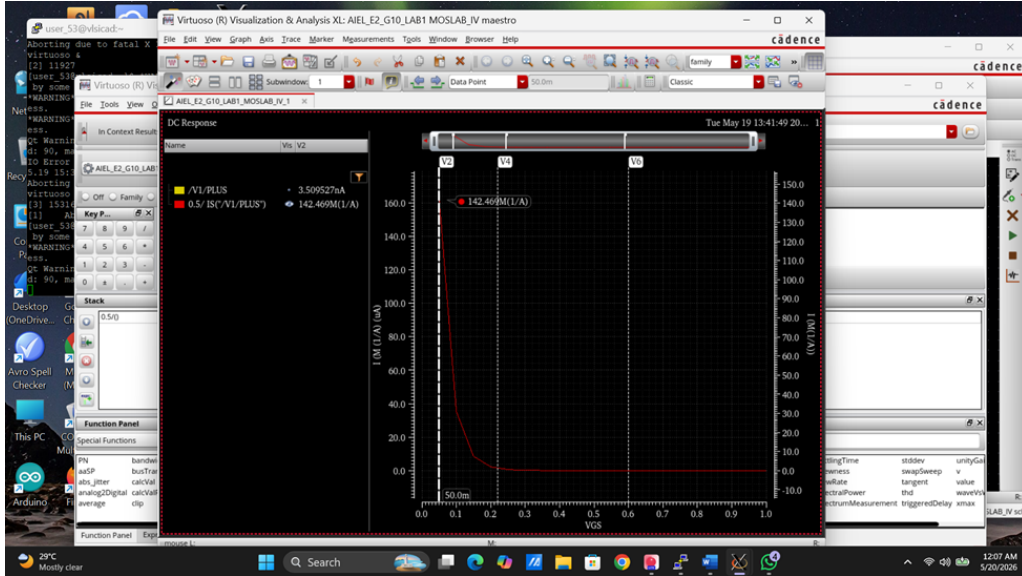


Figure 9: r_o vs V_{GS} curve at $V_{GS} = 50mV$.

From the graph,

$$I_D = 3.509527nA$$

$$r_o = 142.469M\Omega$$

9.3.2 Result at $V_{GS} = 220mV$

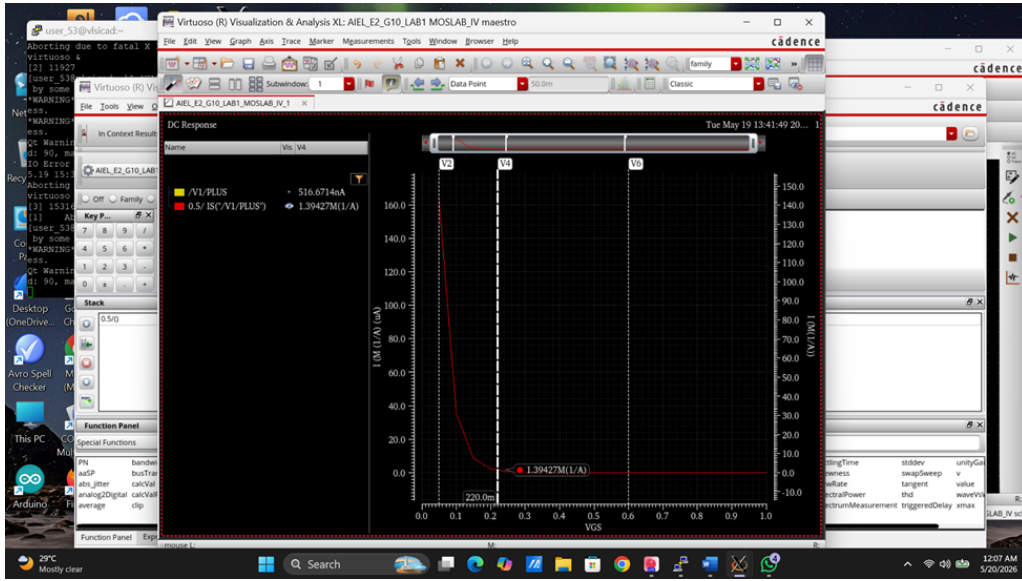


Figure 10: r_o vs V_{GS} curve at $V_{GS} = 220mV$.

From the graph,

$$I_D = 516.6714nA$$

$$r_o = 1.39427M\Omega$$

9.3.3 Result at $V_{GS} = 600mV$

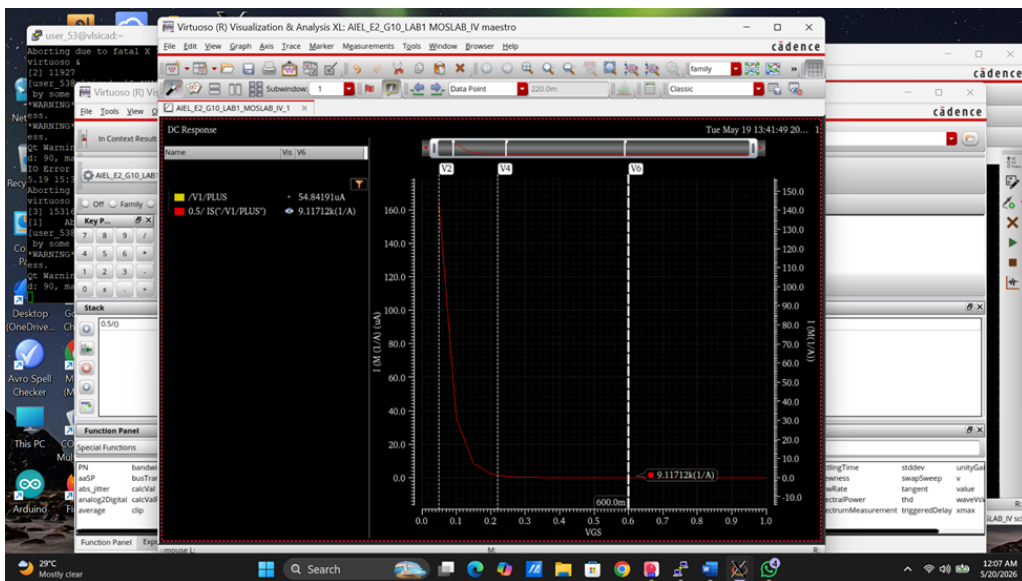


Figure 11: r_o vs V_{GS} curve at $V_{GS} = 600mV$.

From the graph,

$$I_D = 54.84191\mu A$$

$$r_o = 9.11712k\Omega$$

9.3.4 Comment

The r_o vs V_{GS} curve shows that output resistance decreases as V_{GS} increases. At very low V_{GS} , the NMOS transistor is almost in cutoff region and the drain current is extremely small and the output resistance is very high, about $142.469M\Omega$.

When V_{GS} increases to around $220mV$, the transistor starts turning on and enters weak or moderate inversion. The drain current increases, so the output resistance decreases to about $1.39427M\Omega$.

At $V_{GS} = 600mV$, the transistor operates in strong inversion and conducts much higher current. As a result, the output resistance falls significantly to about $9.11712k\Omega$. Therefore, the result confirms that r_o is high in cutoff or weak inversion and becomes much lower as the transistor moves toward strong inversion.

9.4 g_m vs V_{GS} Curve

The transconductance g_m was obtained by differentiating the drain current with respect to gate-to-source voltage:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad (8)$$

9.4.1 Schematic

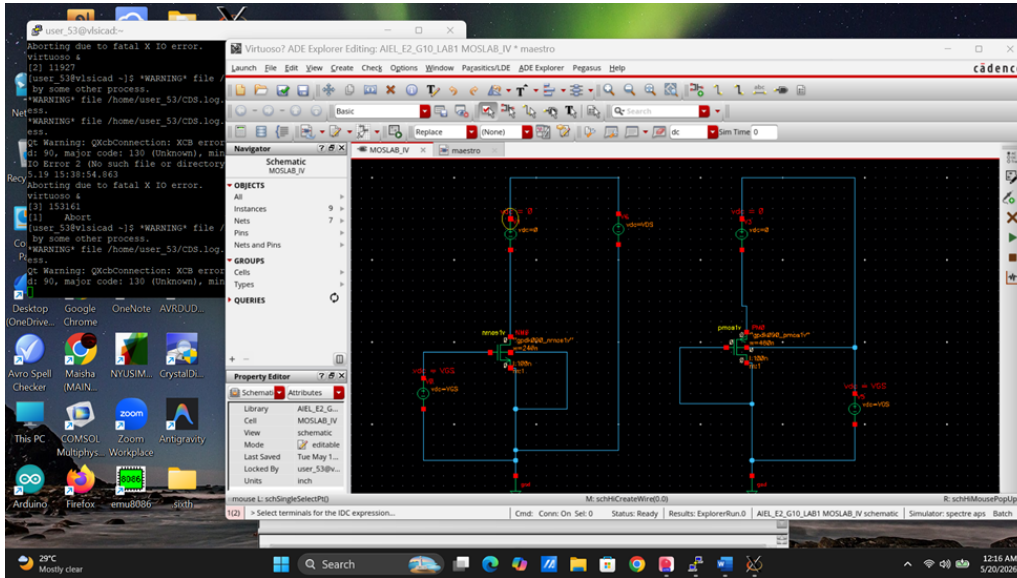


Figure 12: Schematic for g_m vs V_{GS} analysis.

9.4.2 Result



Figure 13: g_m vs V_{GS} curve obtained from derivative of I_D with respect to V_{GS} .

9.4.3 Comment

From the g_m vs V_{GS} curve, it is observed that the transconductance is almost zero at very low V_{GS} because the transistor is in cutoff region and the drain current is negligible. As V_{GS} increases beyond the threshold voltage, the transistor starts conducting and I_D increases rapidly. Since g_m represents the rate of change of drain current with respect to gate voltage, g_m also increases sharply in this region.

After entering strong inversion, the rate of increase of g_m becomes slower and the curve starts to become almost flat. This indicates that although drain current continues to increase with V_{GS} , the incremental change in current per unit change in gate voltage becomes smaller. Therefore, the curve shows that the NMOS transistor has low transconductance in cutoff, rapidly increasing transconductance near threshold, and almost saturated transconductance at higher V_{GS} .

9.5 g_m/I_D vs V_{GS} Curve

The transconductance efficiency was obtained by dividing the transconductance by the drain current:

$$\frac{g_m}{I_D} \quad (9)$$

This parameter is widely used in analog circuit design because it represents the gain obtained per unit drain current.

9.5.1 Result

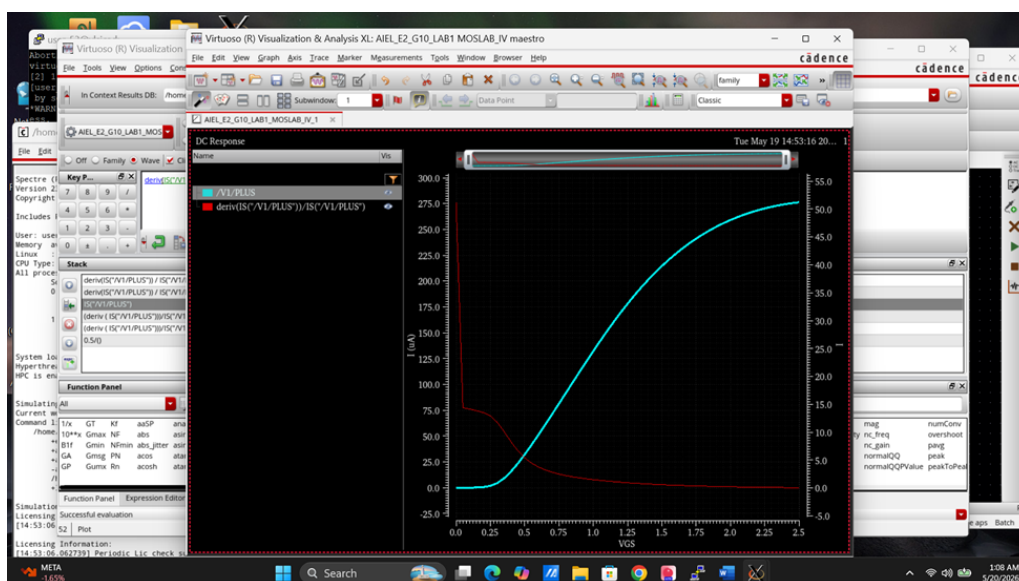


Figure 14: g_m/I_D vs V_{GS} curve of the NMOS transistor.

9.5.2 Comment

From the g_m/I_D vs V_{GS} curve, it is observed that the transconductance efficiency is very high at low values of V_{GS} . In this region, the transistor operates in weak inversion where a small drain current produces relatively large transconductance. Therefore, the value of g_m/I_D becomes maximum.

As V_{GS} increases, the drain current increases rapidly and the transistor gradually moves from weak inversion to moderate and strong inversion. Although g_m continues to

increase, the drain current increases at a much faster rate. As a result, the ratio g_m/I_D decreases continuously with increasing V_{GS} .

At higher gate voltages, the curve becomes almost flat and approaches a very small value. This indicates reduced transconductance efficiency in strong inversion operation. Therefore, the curve confirms that weak inversion operation provides higher gain efficiency and lower power consumption, while strong inversion provides higher current drive but lower efficiency.

10 CONCLUSION

The experiment successfully demonstrated the electrical characteristics of MOS transistors using Cadence Virtuoso simulation. Important device parameters such as threshold voltage, mobility ratio, transconductance, output resistance, and transconductance efficiency were analyzed under different operating conditions.

The results showed that NMOS transistors provide higher carrier mobility and drain current compared to PMOS transistors of equal dimensions. The g_m curve increased with V_{GS} , while the r_o curve decreased as the transistor entered strong inversion. The g_m/I_D characteristic indicated maximum efficiency in weak inversion operation.

Overall, the experiment provided practical understanding of MOSFET operation and helped relate theoretical concepts to real device behavior in integrated circuit design.

11 REFERENCES

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